Ali Abbas

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Education

University of British Columbia September 2023 – April 2027
Vancouver BC

Bachelor of Science in Mathematics, Minor in Data Science - OIS Award

Work Experience

Full-Stack Developer @ Electricity and Water Authority (EWA), Bahrain Jul 2025 – Aug 2025

- Built a full-stack **ticket management system** (Jira-like) with **React**, **Node.js/Express**, and **MongoDB**; designed **RESTful API** endpoints and collections for projects, tickets, and sprints.
- Integrated **Slack API** webhooks for real-time updates and used **OpenAI (GenAI)** to auto-generate project status summaries and stand-up notes.
- Collaborated using **Agile/Scrum** practices (stand-ups, sprint goals) and authored concise **project documentation**/runbooks.
- Deployed frontend to **Vercel** and backend to **Railway**; managed environment variables and monitoring to ensure availability and performance.

Frontend Developer @ UBC Mint Engineering Design Team Sep 2024 – Present

- Developed a **real-time streaming** visualization app in **Next.js** using **WebSockets** to simulate EEG data (256 msg/s) with low-latency charts (**Recharts**) and responsive UI (**shadcn/ui**, **Radix UI**, **Tailwind CSS**).
- Optimized rendering throughput and state updates for **high-frequency data** and contributed to component library patterns for reuse.
- Worked in a cross-functional pod with code reviews (**Git/GitHub**); maintained **documentation** and issue tracking aligned to sprint goals.

Mobile Quality Assurance Engineer @ Qenta | ***Swift, SwiftUI*** July 2024 – August 2024

- Collaborated with cross-functional teams to **debug and resolve app crashes**, logging **over 25 issues** and contributing to **faster release cycles**
- Wrote and maintained **unit and UI tests** using **XCTest** to ensure feature stability, increasing test coverage by
- Employed **concurrency and async/await** for **optimized data handling** and **performance enhancement**.
- Conducted **manual testing** of **iOS** features across **devices and OS versions**, reporting **UI/UX discrepancies**.

Technical Projects

Predictive Maintenance Model | *Python, Power BI*

- Scripted a data processing pipeline in Python using Pandas to clean, structure, and prepare raw industrial sensor data for machine learning analysis.
- Developed a Random Forest model to classify engine health & predict fault conditions based on sensor metrics.
- Designed and deployed an interactive Power BI dashboard connected to model outputs stored in AWS S3, creating a visualization tool to help a maintenance team identify and prioritize high-risk equipment.

Classification Analysis | *R, JupyterLab*

- Utilized R programming and classification techniques to develop predictive models for early diagnosis of heart disease, using datasets from a Heart Disease Database.
- Employed R to select and preprocess relevant columns from the datasets, enabling accurate predictions of heart disease based on a variety of patient factors.
- Utilized KNN to investigate a patient's likelihood of heart disease diagnosis through the analysis of diverse factors.
- Classified the testing set, with an accuracy of about 77.5%, precision of 79.7%, and recall of about 75.0%

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, C++, C#, R, SQL, HTML/CSS, Swift, Racket
Frameworks & Libraries: React, Next.js, Node.js/Express, Tailwind CSS, shadcn/ui, Radix UI, pandas, scikit-learn, Matplotlib, Recharts, WebSockets, XCTest
Data & ML: ETL, data quality/reliability, data warehousing concepts, feature engineering, model evaluation (precision/recall/F1), classification, clustering, visualization, data engineering; **Cloud & DevOps:** Git, GitHub, CI/CD (GitHub Actions), Docker (intro), AWS; **Databases:** MongoDB, SQL (queries, joins), JSON persistence